

Real-Time IoT Patient Monitoring Pipeline using Spark and Dual-Mode Streaming Architecture for Health Analytics.

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Project Title

Real-Time IoT Patient Monitoring Pipeline for Remote Health Analytics: A Dual-Mode Streaming Architecture for Cardiac Risk and Injury Prevention.

Research Team and Contributions

Our team divides responsibilities by the data lifecycle stages:

- **Aiden Le (Data Ingestion):** Responsible for the "front door" of the system. He will build the simulators that generate patient data and manage the messaging system (Kafka) to ensure data arrives correctly and securely.
- **Dasha Lewis (Real-Time Processing):** Responsible for the "brain" of the system. She will write the code that analyzes data as it arrives, calculating health scores and triggering alerts for falls or heart risks.
- **Narasimha Lakavath (Data Quality):** Responsible for "trust and reliability." He will build automated checks to clean the data, ensuring that bad data doesn't trigger false alarms, and create monitoring dashboards to track system health.
- **Pritesh Gurjar (Storage & History):** Responsible for the "memory" of the system. He will design the database architecture that stores historical data for long-term analysis and optimize it for fast reporting.

Objective

Standard remote patient monitoring systems currently face a major limitation: they treat different types of health data as separate silos. Heart monitors (which send data slowly) and fall detection sensors (which send data very fast) are rarely processed together.

Our objective is to build a **novel, unified pipeline** that ingests these conflicting data streams into a single system. We aim to prove that a "Router Pattern" approach can dynamically sort high-speed movement data from low-speed vital signs in real-time. This allows us to detect two distinct risks—cardiac arrest (Sepsis) and physical injury (Falls)—simultaneously, giving doctors a complete picture of patient safety without needing two separate systems.

Materials and methods

To build this system, we will use industry-standard tools and authentic datasets:

Data Sources: We are using two open-source datasets to simulate a realistic environment:

1. **WISDM Dataset:** Contains real accelerometer data from people walking and jogging. We will use this to simulate "Fall Detection" sensors.
2. **Kaggle Health Dataset:** Contains heart rate and sleep metrics. We will use this to simulate "Vital Signs" monitors.

Methodology (The "Router" Approach): Our system functions like a traffic controller. As data arrives, our processing engine reads the device tag.

- If it is **Movement Data**, it is routed to the "Safety Logic," which checks for sudden impacts (Falls).
- If it is **Vitals Data**, it is routed to the "Health Logic," which calculates a Modified Early Warning Score (MEWS) for sepsis risk.

Storage Strategy: We will organize data into three layers: **Raw** (original data), **Cleaned** (standardized and checked for errors), and **Curated** (summarized for daily reports).

Expected Outcomes

By the end of this project, we expect to deliver:

1. **A Working End-to-End Pipeline:** A system that takes in raw data and produces actionable insights without manual intervention.
2. **Real-Time Alerts:** The system will trigger immediate notifications when a simulated patient falls or their heart rate enters a critical zone.
3. **Unified Dashboard:** A visual interface showing both current threats (alerts) and historical trends (daily health summaries).
4. **Performance Benchmarks:** We will prove the system is fast enough for healthcare by measuring the time it takes to go from "Event Occurred" to "Alert Sent" (Target: under 2 seconds).

Discussion

This project addresses a critical gap in modern healthcare engineering. While devices are getting smarter, the infrastructure to handle their data remains fragmented. By successfully combining high-frequency sensor data with clinical vital signs, we are demonstrating a scalable blueprint for the next generation of "Smart Hospitals." Our approach highlights the importance of data quality and efficient routing over simple storage, ensuring that life-saving alerts are prioritized over general logging.

Conclusion

We propose a robust, achievable data engineering project that solves the problem of disconnected health data. By integrating two distinct datasets into one dual-mode analysis pipeline, we will demonstrate how modern streaming technologies can improve patient safety. Our team has a clear division of labor covering ingestion, processing, quality, and storage, ensuring a successful delivery of a production-grade prototype.